

10s10p Battery Pack Fault & Mitigation

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ABSTRACT

This project focuses on the modeling, fault detection, and mitigation of a 10S-NP lithium-ion battery pack using MATLAB/Simulink. The battery pack was developed using predefined cell parameters based on the Molicel P45B cell model. The study was carried out to understand how faults affect battery performance and to explore suitable methods to improve the safety and reliability of battery systems used in electric vehicle applications.

An internal resistance increase fault was introduced into one of the cells to analyze its impact on the overall pack behavior. Parameters such as current, voltage, and temperature were monitored during simulation. Temperature comparison between the affected cell and healthy cells was used to study the thermal response of the system and identify abnormal operating conditions. Fault detection was implemented using threshold-based monitoring, where cell temperature and current values were compared against predefined limits.

Different mitigation strategies were investigated to reduce the effects of battery faults. Overcurrent protection was implemented using a fuse model to simulate automatic circuit interruption under unsafe current conditions. A MOSFET-based disconnect mechanism was also developed to isolate the battery pack when a fault condition was detected. In addition, bypass path and current limiting approaches were considered as protection methods for maintaining system operation and reducing stress on the battery pack.

The simulation results demonstrate the importance of continuous monitoring and fault mitigation in battery systems. The developed model provides a practical approach for studying battery faults and protection techniques, and highlights how simulation tools can be used to evaluate battery safety strategies before real-world implementation.

Tools and Resources Used

The project was developed primarily using **MATLAB Simulink and Simscape**, which were used for battery pack modeling, fault implementation, thermal analysis, simulation, and mitigation strategy development. The battery model was created using predefined cell parameters and simulation blocks available within the MATLAB environment.

ChatGPT was used as a supporting tool for understanding complex concepts related to battery fault modeling, MATLAB/Simulink block implementation, fault detection logic, protection methods such as fuse and MOSFET disconnect systems, troubleshooting simulation issues, and improving technical documentation.

Google Search and online technical resources were used to refer to MATLAB documentation, battery modeling concepts, electrical protection methods, and general background information related to lithium-ion battery systems and fault mitigation strategies.

Additional references included **MATLAB documentation**, library browser resources, and online tutorials for understanding specific block functions, parameter settings, and simulation workflows

ChatGPT was used during different stages of this project to support learning, problem solving, and documentation. Assistance from ChatGPT was mainly used to understand MATLAB/Simulink concepts related to battery modeling, fault implementation, and mitigation strategies.

ChatGPT was used to gain clarity on the implementation of battery fault conditions such as internal resistance increase and to understand their effects on current flow, voltage behavior, and temperature variation within the battery pack. Support was also taken in understanding the setup and working of protection and mitigation mechanisms including fuse protection, MOSFET disconnect systems, bypass paths, and current limiting methods in MATLAB/Simulink.

In addition, ChatGPT was used for troubleshooting simulation issues, interpreting simulation results, understanding block configurations, and resolving signal connection methods between Simulink and Simscape components. It was also used to assist in improving technical explanations, organizing the report structure, and drafting sections such as the abstract and system description.

All modeling, simulation setup, parameter selection, testing, and final validation of results were carried out by the author. ChatGPT was used as a supporting tool for technical understanding, debugging assistance, and documentation preparation.

Cell Parameters and Battery Model

The battery pack model used in this project was developed in **MATLAB/Simulink** using the predefined lithium-ion cell parameters available within the MATLAB battery library. The selected cell model was based on the **Molicel P42A** lithium-ion cell, which provides electrical and thermal characteristics suitable for high-performance battery applications.

The predefined cell model includes important parameters such as **nominal voltage, rated capacity, internal resistance, open-circuit voltage (OCV) characteristics, state-of-charge (SOC) behavior, thermal properties, and charge-discharge performance data**. The Molicel P42A cell has a nominal voltage of approximately **3.6–3.7 V**, a nominal capacity of **4.2 Ah**, and is designed for high current operation, making it suitable for electric vehicle and performance-oriented battery applications.

These predefined parameters were used as the base for constructing the **10S-NP battery pack model**. The model incorporates the electrical behavior of the cell along with thermal effects, enabling the study of temperature variation, current flow, voltage response, and fault behavior under different operating conditions. Using a predefined cell model allowed realistic simulation results while reducing the need for manual parameter estimation and characterization.

The battery model formed the foundation for implementing fault conditions such as **internal resistance increase**, and for evaluating protection and mitigation methods including **fuse protection, MOSFET disconnect mechanisms, bypass paths, and current limiting strategies**.

Block Parameterization Manager: Battery (Table-Based)

SELECT

FORMAT

Apply all

Reset all

ManufacturerMolice

PARAMETERIZE

FILTER

Select part

Part specification

Manufacturer	Part number	Battery type	Capacity (mA*hr)	Vnominal (V)	Weight (g)	Geometry
Molice	INR_21700_P45B	Lithium-ion	4500	3.6000	70	Cylindrical

Attribute	Value
Manufacturer	Molice
Part number	INR_21700_P45B
Part series	
Part type	Battery (Table-Based)
Date part added	15-Jun-2023
Parameterization note	Predefined parameterizations of Simscape compone...
Collection ID	MW.SimscapeBatteryShared

Compare selected part with block

Parameter name	Part value origin	Unmatched values	Part value:INR_21700_P45B	Current block value	Unit
Main>Vector of state-of-charge values, SOC	Datasheet derived	☒	[0 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.1 0...	[0 0.1 0.25 0.5 0.75 0.9 1]	1
Main>Vector of temperatures, T	Datasheet derived	☒	[233.15 243.15 253.15 273.15 296.15 318.15 333.15]	[278 293 313]	K
Main>Open-circuit voltage, V0(SOC,T)	Datasheet derived	☒	[3.17242 3.17242 3.17242 3.17242 3.17242 3.17242 ...	[3.49 3.5 3.51;3.55 3.57 3.56;3.62 3.63 3.64;3.71 3.7...	V
Main>Open-circuit voltage, V0(SOC)	Datasheet derived	☒	[3.17242 3.18272 3.19302 3.20331 3.21361 3.22391...	[3.5057 3.566 3.6337 3.7127 3.9259 4.0777 4.1928]	V
Main>Terminal voltage operating range [Min Max]	Datasheet derived	☒	[2.25 Inf]	[0 Inf]	V
Main>Terminal resistance, R0(SOC,T)	Datasheet derived	☒	[0.3026 0.35142 0.34349 0.24415 0.02095 0.15874 ...	[0.0117 0.0085 0.009;0.011 0.0085 0.009;0.0114 0.0...	Ohm
Main>Terminal resistance, R0(SOC)	Datasheet derived	☒	[0.02095 0.01921 0.01774 0.0166 0.01557 0.01469 ...	[0.0085 0.0085 0.0087 0.0082 0.0083 0.0085 0.0085]	Ohm
Main>Terminal resistance during discharging, R0(SO...	Parameter not set	☐	[0.0117 0.0085 0.009;0.011 0.0085 0.009;0.0114 0.0...	[0.0117 0.0085 0.009;0.011 0.0085 0.009;0.0114 0.0...	Ohm
Main>Terminal resistance during discharging, R0(SOC)	Parameter not set	☐	[0.0085 0.0085 0.0087 0.0082 0.0083 0.0085 0.0085]	[0.0085 0.0085 0.0087 0.0082 0.0083 0.0085 0.0085]	Ohm

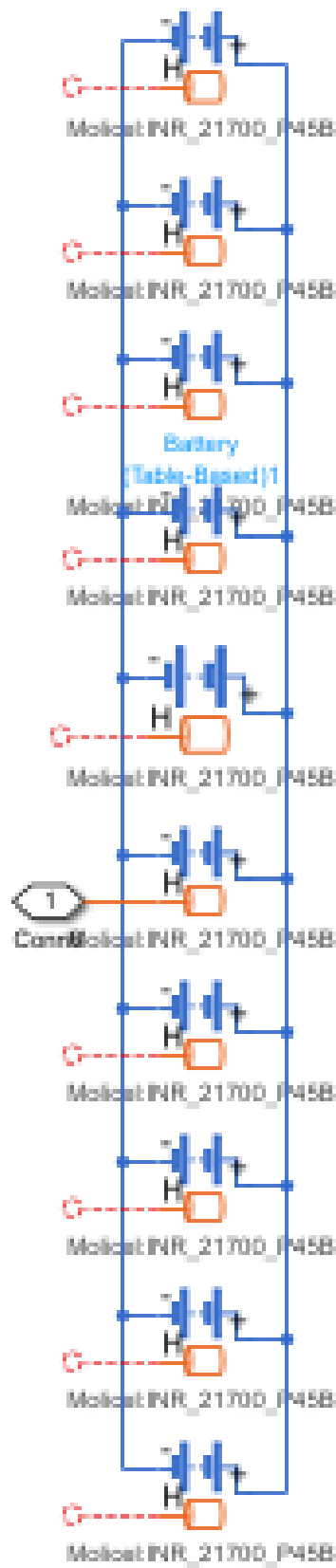
Parallel Cell Configuration

The battery pack model was constructed by connecting predefined **Molice P42A cells in a parallel configuration**, where multiple identical cells were grouped together to form a parallel cell unit. In this configuration, the cells share a **common positive terminal, common negative terminal, and a common thermal connection point**. The parallel arrangement allows the cells to operate as a single unit with increased current carrying capability and improved overall capacity.

Each parallel group was created using the predefined battery cell blocks available in MATLAB/Simulink. Since the cells are connected in parallel, the **voltage across all cells remains the same**, while the total capacity and current capability increase based on the number of cells connected. This configuration is commonly used in electric vehicle battery systems to meet higher power and energy requirements.

The model includes a **single thermal node connection** for the parallel group to represent heat generation and thermal behavior within the cell assembly. The thermal port enables temperature monitoring and thermal interaction during charging, discharging, and fault conditions. Common **positive and negative electrical terminals** are used to simplify the representation of current distribution within the parallel network while maintaining realistic pack behavior.

The parallel cell model forms the basic building block of the overall battery pack. These parallel groups are later connected in series to achieve the required pack voltage. The configuration allows the simulation of electrical and thermal effects such as current sharing, temperature variation, and fault propagation within the battery system. This arrangement also provides a suitable platform for studying fault detection and mitigation strategies under different operating conditions.



Battery Pack Configuration

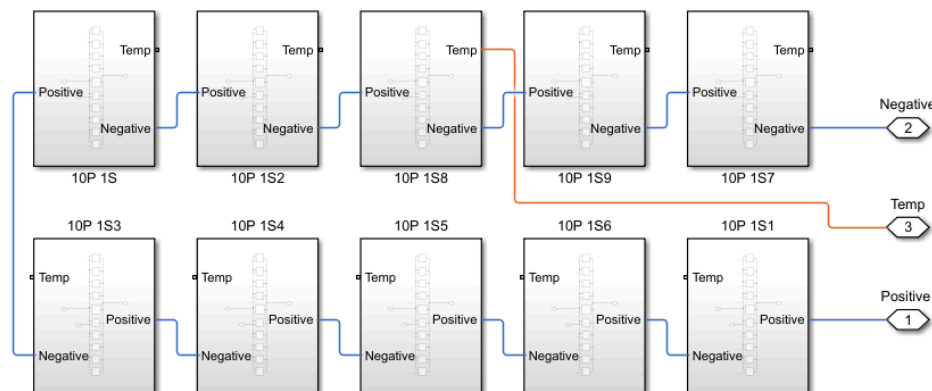
The battery pack model was developed using a **10S10P configuration** in MATLAB/Simulink. Initially, predefined **Molicel P42A cell models** were connected in parallel to form a parallel cell group. Each parallel group consisted of **10 cells connected in parallel**, allowing an increase in total capacity and current carrying capability while maintaining the same voltage across the group.

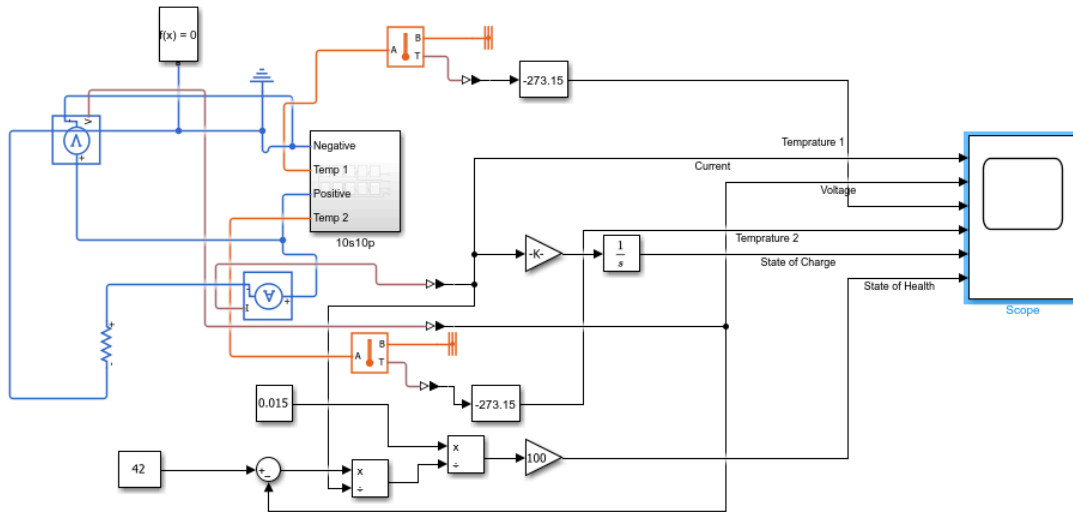
These parallel cells shared a **common positive terminal, common negative terminal, and a thermal connection point**, enabling both electrical and thermal behavior to be represented within the model. The use of parallel grouping helped simulate realistic battery behavior under load conditions, including current distribution and temperature response.

After creating the parallel cell groups, **10 such groups were connected in series** to achieve the required pack voltage, resulting in the final **10S10P battery pack configuration**. The series connection increases the overall output voltage of the pack while maintaining the capacity provided by each parallel group.

To improve model organization, readability, and ease of modification, each stage of the battery model was structured using **subsystems within Simulink**. Individual cells were grouped into parallel modules, and the parallel modules were further arranged into higher-level subsystems before being assembled into the final battery pack configuration. This modular approach simplified connections, improved simulation management, and made it easier to implement fault conditions, sensing elements, and mitigation strategies at different levels of the battery pack.

The subsystem-based architecture also allowed easier integration of monitoring components such as temperature sensing, fault detection logic, and protection mechanisms including fuse protection, MOSFET disconnect systems, bypass paths, and current limiting methods. This structured modeling approach supported efficient development and analysis of the complete battery pack system.





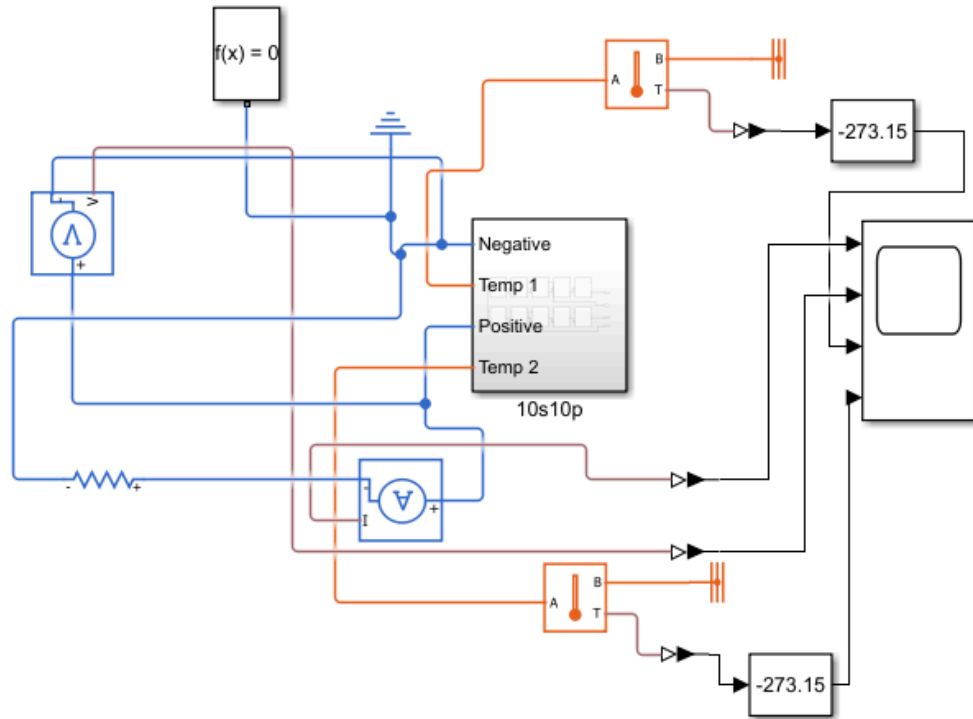
Final Simulink Model

Simulation Results

After completing the modeling, fault implementation, and mitigation setup, simulations were carried out in MATLAB/Simulink to observe the behavior of the battery pack under different operating conditions. The outputs obtained from the simulations included parameters such as **current, voltage, temperature variation, and fault response** of the battery system.

The simulation results were analyzed to study the effect of the introduced fault conditions and the response of the implemented mitigation strategies. Changes in temperature, current flow, and system behavior before and after fault occurrence were observed and compared.

The corresponding **simulation graphs, waveform outputs, and circuit diagrams** generated during the analysis are attached in the following sections to illustrate the battery pack configuration, fault behavior, and protection mechanism performance. These outputs provide a visual representation of the system operation and help validate the implemented model and mitigation strategies.



The battery parameters were monitored using sensing components available in MATLAB/Simulink. Temperature variations within the battery pack were measured using **thermal/temperature sensors** connected to the thermal ports of the cells, allowing the thermal behavior of both healthy and faulted cells to be observed during simulation. The **voltage sensor** was used to measure the voltage output and voltage changes across the cells and battery pack, helping identify the effect of the introduced fault on electrical performance. Similarly, a **current sensor** was used to monitor the current flow through the system, enabling analysis of current behavior under normal and fault conditions. These measured parameters were used to evaluate the effect of the internal resistance fault and the response of the implemented mitigation strategies.

The **State of Health (SOH)** of the battery pack was estimated using battery monitoring approaches based on **Coulomb counting** and **internal resistance analysis**. Coulomb counting was used to track the charge flowing into and out of the battery over time, providing an indication of the available capacity compared to the expected battery performance. Changes in the effective charge handling capability of the battery were used to understand the level of degradation occurring within the system.

In addition to this, SOH estimation was also related to the variation in **internal resistance** of the cell. As batteries age or develop faults, their internal resistance typically increases, leading to greater power losses, voltage drops, and reduced overall performance. By comparing the resistance behavior of the faulted cell with the expected healthy cell characteristics, the condition of the battery could be assessed. The combined use of charge monitoring and resistance-based evaluation provided a theoretical

understanding of battery health and helped identify the impact of the introduced internal resistance fault on the system.



Graph Before Fault

Internal Resistance Fault Implementation and Analysis

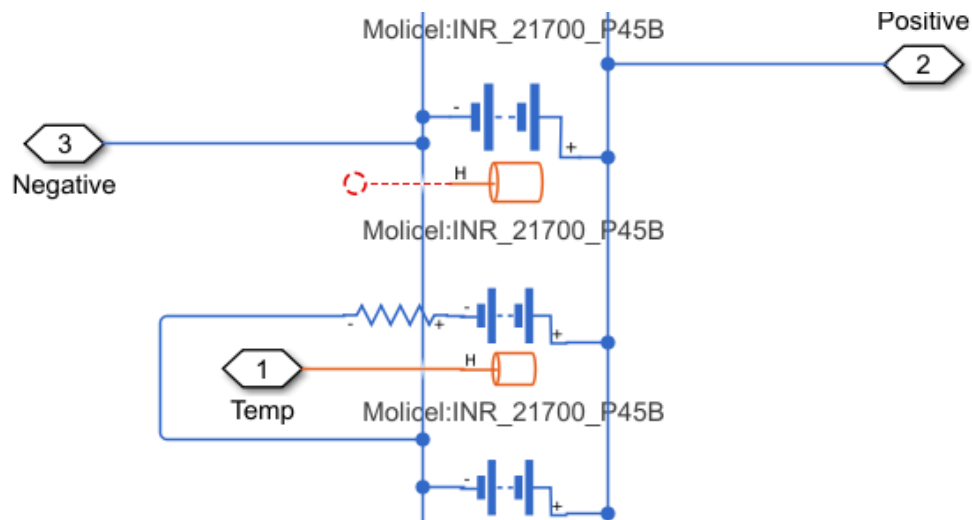
An **internal resistance fault** was implemented in the battery pack model by introducing an **additional resistor in series with one of the cells** within the configuration. This was done to represent the effect of increased internal resistance that may occur in real battery systems due to aging, degradation, poor connections, or internal cell damage.

After introducing the fault, noticeable changes were observed in the electrical and thermal behavior of the affected cell and the overall battery pack. One of the main observations was a **variation in cell temperature**, indicating altered power dissipation and energy loss within the faulted branch. The change in resistance affected the current flow and resulted in modified thermal behavior compared to the healthy cells in the pack.

A **reduction in the State of Health (SOH)** of the affected cell was also observed. The increased resistance negatively impacted the effective performance of the cell, representing the behavior of a degraded battery condition. In addition to the SOH reduction, a **drop in voltage output** was noted due to the additional voltage loss across the introduced resistance. This affected the electrical performance of the battery pack and highlighted the influence of internal resistance on pack efficiency and stability.

Other related parameters such as current distribution, voltage response, and thermal characteristics also showed measurable changes after the fault was introduced. The observed results demonstrate how an

increase in internal resistance can influence battery operation and emphasize the importance of fault detection and mitigation methods in maintaining battery reliability and safety.



After the internal resistance fault was introduced by adding an external resistor in series with one of the cells, a change in the temperature measured by the **thermal sensor attached to the faulted cell** was observed. The temperature of the monitored cell decreased instead of increasing. This behavior can be attributed to the additional series resistor altering the current flow and sharing the power dissipation within the circuit. A portion of the energy loss and heat generation shifted to the added resistor, reducing the thermal loading experienced directly by the cell being monitored.

Although the added resistor is expected to dissipate heat during operation, a direct temperature measurement of the resistor was not implemented in the model because no dedicated thermal sensing arrangement was attached to that component in the simulation setup. Therefore, the thermal analysis was based primarily on the temperature readings obtained from the battery cell thermal sensors. The observed behavior indicates that the introduced resistance affected both the electrical and thermal characteristics of the system, influencing parameters such as current flow, voltage response, and measured cell temperature under fault conditions.